

DAY-1

Difference Between Web Application and Web Service (Short)

Web Application

- A web application is a program that runs on a web server and is accessed through a browser.
- It provides a user-friendly interface for interaction.
- Users can perform tasks like logging in, shopping, or sending emails.

Example:

- Online shopping websites
- Internet banking
- Student portal

Flow:

User → Browser → Web Server → Database

Web Service

- A web service is an API that enables different applications to communicate over the internet.
- It does not have a user interface.
- It simply receives requests, processes them, and returns data.

Example:

- Payment processing API
- Weather API
- Currency exchange API

Flow:

Application A → Web Service → Application B

What does Spring Boot do?

- Handles client requests.
- Implements business logic.
- Connects to the database.
- Provides REST APIs (web services).
- Returns data (usually JSON) to the frontend.

So, when you're using **Spring Boot** in the backend, you're building a **Java-based backend application** that can act as:

- a **backend for a web application**, and/or
- a **web service (REST API)** that other applications can consume.

In short: **Spring Boot is a Java-based backend framework used to develop web applications and RESTful web services.**

SQL (Structured Query Language)

SQL (Structured Query Language) is a standard language used to **store, retrieve, update, and manage data** in a relational database.

Types of SQL Commands

1. **DDL (Data Definition Language)** – Defines the database structure. Has auto commit
 - **CREATE** – Creates a table or database.
 - **ALTER** – Modifies a table.
 - **DROP** – Deletes a table or database.
 - **TRUNCATE** – Removes all records from a table.
2. **DML (Data Manipulation Language)** – Manipulates data. No auto commit
 - **INSERT** – Adds new records.
 - **UPDATE** – Modifies existing records.
 - **DELETE** – Removes records.
3. **DQL (Data Query Language)** – Retrieves data.
 - **SELECT** – Fetches data from a table.
4. **DCL (Data Control Language)** – Controls user permissions.
 - **GRANT** – Gives privileges.
 - **REVOKE** – Removes privileges.
5. **TCL (Transaction Control Language)** – Manages transactions.
 - **COMMIT** – Saves changes permanently.
 - **ROLLBACK** – Undoes changes.

- **SAVEPOINT** – Creates a point to roll back to.

Difference Between **VARCHAR** and **VARCHAR2**

VARCHAR	VARCHAR2
ANSI SQL standard data type.	Oracle-specific data type.
Behavior may change in future Oracle versions.	Recommended and commonly used in Oracle.
Stores variable-length character data.	Also stores variable-length character data.
Rarely used in Oracle.	Widely used in Oracle databases

Default Values in SQL

A **DEFAULT value** is a value that is automatically assigned to a column if no value is provided during an **INSERT** operation.

Syntax

```
CREATE TABLE Employee (
  EmpID INT,
  Name VARCHAR(50),
  City VARCHAR(30) DEFAULT 'Hyderabad'
);
```

Example

```
INSERT INTO Employee (EmpID, Name)
VALUES (101, 'Ashritha');
```

Result:

EmpID	Name	City
101	Ashritha	Hyderabad

Since no value was given for **City**, the **default value** '**Hyderabad**' is inserted automatically.

Advantages

- Automatically assigns a value when none is provided.
- Reduces **NULL** values.

- Ensures data consistency.

EXAMPLES:

For null values:

```
select eid,ename from tbl_emp where ename is null;
```

For not null values:

```
select eid,ename from tbl_emp where ename is not null;  
Other then above other operators won't work for null
```

To get that values in the records:

```
select * from tbl_emp where eid in (101,103);
```

To get other then that values:

```
select * from tbl_emp where eid in (101,103);
```

Ignorance of unknown value:

```
select * from tbl_emp where eid not in (101,105);
```

Between the values:

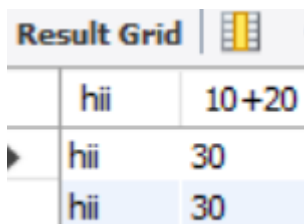
```
select * from tbl_emp where eid between 101 and 103;
```

Between value with higher limit and lower limit (it ignore so,no output):

```
select * from tbl_emp where eid between 103 and 101;(101-103=gives -ve val)
```

We can pass any literal after select query ,we get that literal repeated according to the number of records in the table:

```
select "hii",10+20 from tbl_emp;
```



Result Grid	
hii	10+20
hii	30
hii	30

O/p:

sysdate() is function in mysql but it is variable in oracle:

```
select eid,sysdate(),"hii",10+20 from tbl_emp;
```

By using alias “as” we can change display name:

```
select eid as id,sysdate(),"hii",10+20 from tbl_emp;
```

We use double quote for two words in alias:

```
select eid as id,sysdate() as "Curr date","hii",10+20 from tbl_emp;
```

Add new column:(to assign values for new column we use update query)

```
alter table tbl_emp add column dno int;
```

Drop column:

```
alter table tbl_emp drop column dno;
```

Rename column:

```
alter table tbl_emp rename column eid to empid;
```

Truncate table:

```
truncate table tbl_emp;
```

Rename Table:

```
Rename table tbl_emp to tbl_emp1;
```

Drop table:

```
Drop table tbl_emp1;
```